

DES Encryption and Decryption Engine

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Abstract: We designed a Data Encryption Standard (DES) engine with five inputs and one output, utilizing ModelSim, Cadence Innovus, and Cadence Genus. We ensured that the chip had no errors by performing DRC and LVS at every step and achieved a positive slack, resulting in a stable operating frequency and sufficient timing margin. In this report, we discuss the reasons behind designing a DES engine, the chip design process steps involved, the verification process, and the chip's performance.

Introduction: Hardware cryptography is essential for most modern IoT devices, communications, and computing platforms. Although DES is no longer as commonly used as Advanced Encryption Standard (AES) or other more modern encryption schemes, the experience from designing a DES chip can be applied to understand different types of block ciphers. Additionally, the DES algorithm is well-documented, and numerous known test vectors are available, making it easy for us to test all stages of the algorithm while writing the Verilog code and to verify the functional correctness of our final synthesized design. Due to its relevance, educational value, and efficiency in testing, we chose to create a DES chip with an SPI for our final project.

Overview: This work describes a hardware implementation of the DES cryptographic engine fabricated using 180nm CMOS technology. DES is a block cipher that processes 64-bit data blocks through an iterative procedure [1]. The design supports both encryption and decryption modes, selectable via a mode-select signal. The architecture utilizes a 16-round Feistel structure, a dedicated key expansion unit, eight parallel S-boxes [1], and a control finite state machine. An SPI (Serial Peripheral Interface) serves as

the communication protocol for transferring keys, selecting encryption/decryption modes, plaintext, and ciphertext. The Verilog code specifying the DES core and SPI behavior is synthesized to produce a final chip that adheres to TSMC design rules and the specified area constraint of 2mm x 2mm.

Chip Process Steps: The chip design process began with RTL freeze, where we implemented and verified all RTL Verilog modules using ModelSim, as shown in Fig. 1. After confirming functional correctness, we pre-synthesized the RTL with Cadence Genus and validated the resulting gate-level netlist through the top-level gate-level simulations. We then developed a top-level wrapper for the top-level module, including the I/O pad interface. We performed post-synthesis simulations on the top-level wrapper with Genus, obtaining a gate-level netlist and an SDC file. Using the gate-level netlist and SDC constraints, we performed floorplanning, power planning, clock tree synthesis, and routing. We added the filler cells in Innovus, verifying that there were no geometry or antenna violations, as shown in Fig. 3. The layout and netlist were imported into Virtuoso for DRC and LVS checks at the chip level; once clean, we inserted the sealring and revalidated DRC and LVS in Fig. 4 and Fig. 5. After that, we added polyfill and metalfill, and performed another full-chip DRC and LVS before adding the VDD and VSS pads and wiring the sealring and VSS, as shown in Fig. 6 and Fig. 7. We added M6 wire to connect them, and the final DRC/LVS verification passed.

RTL Verification: The functionality of the DES engine that we implemented in Verilog was verified using ModelSim. We started with testbenches for individual modules: the S-box, F-function, initial permutation, final permutation, SPI, PC1, PC2, key scheduler, and the control state machine. For all the submodules, we first used known DES test vectors, such as an all-zero input. Then, we used a spreadsheet that calculates all intermediate values in the DES algorithm to verify that the results of each submodule were correct with custom plaintext inputs [2]. The S-box array module comprises eight S-box submodules. The S-box

testbench verifies that all eight S-boxes produce the expected results and are combined properly in the S-box array module. We validated the F-function with its testbench, comparing intermediate expansion, S-box, and P-box outputs with the expected values. The control state machine testbench verified proper FSM sequencing across the 16-round Feistel structure. The key schedule module was validated to correctly generate all 16 round subkeys from the input key, again using the intermediate and final key values calculated from the spreadsheet [2].

ModelSim Simulation - Pre-Synthesis: After verifying that all individual submodules functioned correctly, we tested the integrated top module. The top module testbench verified the four-transaction SPI protocol (KEY, DATA, CONTROL, WAIT) by sending 64-bit key and plaintext values serially over MOSI and reading the resulting ciphertext on MISO, as shown in Fig. 1. Test cases for the top module included known DES vectors, custom patterns from the spreadsheet, and edge cases. We also verified each encryption result by performing the corresponding decryption to recover the original plaintext. All testbenches produced outputs that matched the expected values, confirming the functional correctness of the RTL design before synthesis.

ModelSim Simulation - Post-Synthesis: The post-synthesis simulation was performed in ModelSim using the gate-level netlist with the RTL-freeze testbench to ensure that no behavioral changes occurred during synthesis. The netlist produced clean and stable waveforms, and the post-synthesis simulation completed successfully with no timing violations or functional discrepancies, as shown in Fig. 2.

DRC and LVS: We performed DRC verification using Cadence Innovus with TSMC 180nm design rules. It checked all geometric constraints across the six metal layers, polysilicon, and active regions [3]. Also, it examined minimum width, spacing, enclosure, and density requirements for all layout layers. After

completing the place-and-route flow, including clock tree synthesis, filler cell insertion, and final optimization, the DRC analysis reported zero violations across all layers. We did LVS verification to ensure complete equivalence between the physical layout and the original circuit netlist. To compare with the extracted netlist from the layout geometry and the gate-level netlist from synthesis, the hierarchical LVS checked all 23,702 instances. The clean DRC and LVS were verified before sealring named as top_pad layer, and after sealring named as top_pad_merged layer in the Virtuoso tool. Fig. 4 and Fig. 5 provided zero violations in design rules and matched the physical layout with the netlist aspect. After adding polyfill and metalfill to the initial metal chip layer, the clean DRC and LVS results were satisfactory, as shown in Fig. 6 and Fig. 7.

Performance: Having verified that the chip is functionally correct and has no violations, we report on its performance in this section. The design achieved a stable 33.3 MHz operating frequency and a timing margin of approximately +4.315 ns. Power distribution across the functional blocks was kept uniform through careful routing and placement adjustments. The power distribution network utilizes a hierarchical grid that spans all six metal layers.

The synthesized netlist clusters into functional regions based on connectivity. The implementation primarily uses NAND2X1 and NAND3X1 gates, with 56 buffers inserted along high-fanout nets. Metal layer allocation optimizes signal integrity and routing efficiency. M1 only handles intra-cell connections and local power rails. M2-M4 carry signal routing: M2 for vertical local connection; M3 for the majority routing, including permutations; M4 for overflow and critical paths. M5/M6 exclusively serves power distribution. As a result, the routing process was completed with no overflow, and track utilization remained well-balanced across the layout. The compact pad ring contains 236 structures on the 2.607 mm² die perimeter. We implemented five inputs, cs_n, rst, sclk, mosi, and clk, and one output, miso. Power delivery uses a VDD

and a GND pad sized for 8.87 mW. Four corner pads provide mechanical stability and sealing continuity, while 224 filler pads maintain ring integrity and moisture protection. Since SPI pads are grouped consecutively to minimize skew, clock and reset inputs sit on opposite die sides to reduce coupling.

The slack results from RTL synthesis are summarized as follows. For the core TopModule, the initial generic synthesis step reported a slack of -354 ps, which improved to 0 ps after logic mapping, optimization, and final synthesis. For the top_pad design, including the I/O pad frame, the generic synthesis step showed a positive slack of +3,617 ps. After mapping, optimization, and final synthesis, the final slack converged to +362 ps. Before Clock Tree Synthesis (CTS), the chip had a timing with a WNS of -12.742 ns. The synthesis tool assumed an ideal zero-skew clock distribution; however, physical routing introduces delays, resulting in timing violations at the target frequency of 33.3 MHz (30 ns period). CTS resolved all violations by implementing a balanced H-tree topology. Post-CTS timing achieved +4.315 ns WNS with zero failing paths. The clock network consumes only 0.319 mW, efficiently enabling timing closure without requiring manual redesign.

Conclusion: We completed the ASIC design flow from RTL to GDSII by implementing a 64-bit DES processor. Through Verilog, the design achieved 100% verification coverage across all modules using standard DES examples. The physical implementation has 4,524 gates operating at 33.3 MHz within a 2.295 mm² die area. The final layout, “top_pad_merged”, passed all requirements with zero DRC/LVS violations across 2,847 design rules, which achieved positive timing slack after clock tree synthesis. Thus, our comprehensive chip design flow demonstrates proficiency in digital design methodology and the preparation of fabrication-ready tape-outs.

References:

[1] GeeksforGeeks, "Data Encryption Standard (DES): Set 1," *GeeksforGeeks*, <https://www.geeksforgeeks.org/computer-networks/data-encryption-standard-des-set-1/> (accessed Dec. 2, 2025).

[2] "DES cipher internals in Excel," *Project Nayuki*, <https://www.nayuki.io/page/des-cipher-internals-in-excel> (accessed Dec. 2, 2025).

[3] "Cadence Physical Verification System Datasheet," *Cadence*, 2025. https://www.cadence.com/en_US/home/resources/datasheets/cadence-physical-verification-system-ds.html (accessed Dec. 2, 2025).

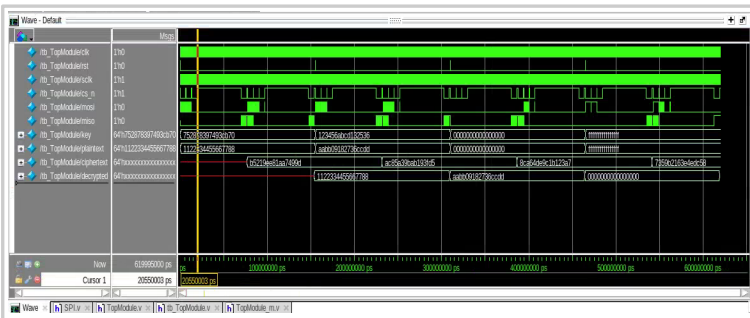


Fig. 1: Pre-synthesis ModelSim simulation on the top-level module



Fig. 2: Post-synthesis ModelSim simulation on the gate-level netlist

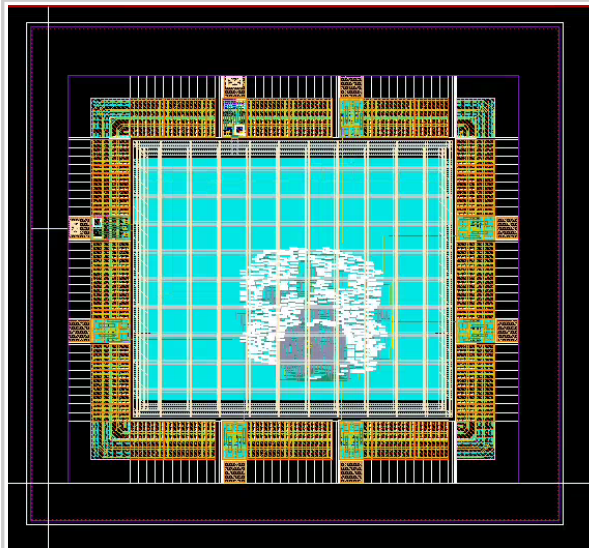


Fig. 3: Full-chip view

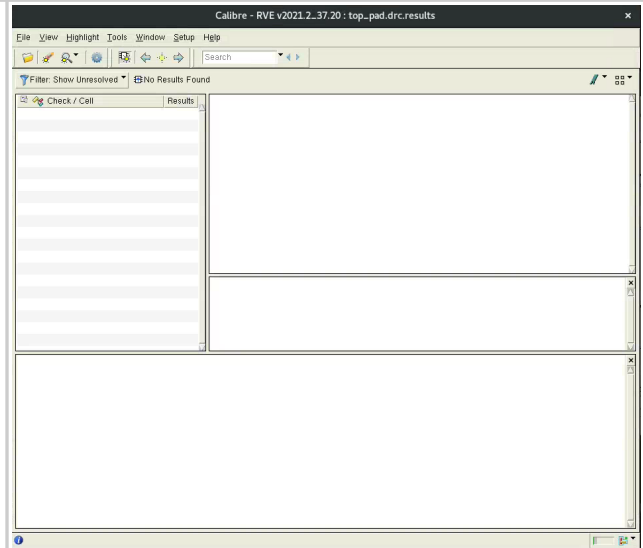


Fig. 4: The clean DRC: pre-sealing

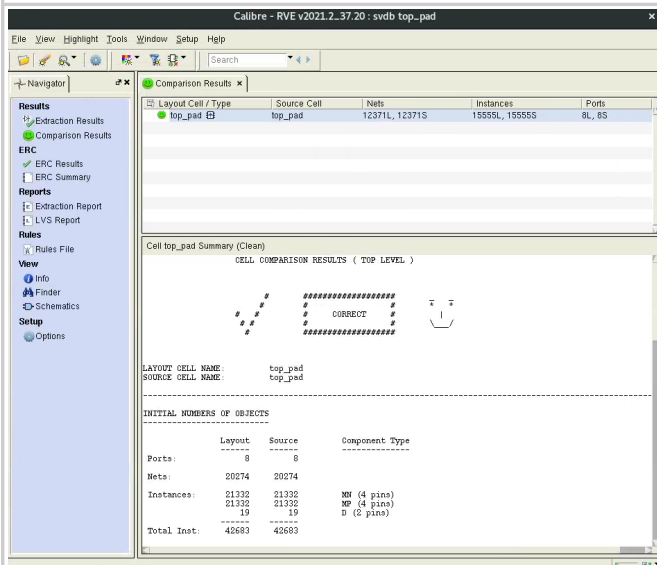


Fig. 5: The clean LVS: pre-sealing

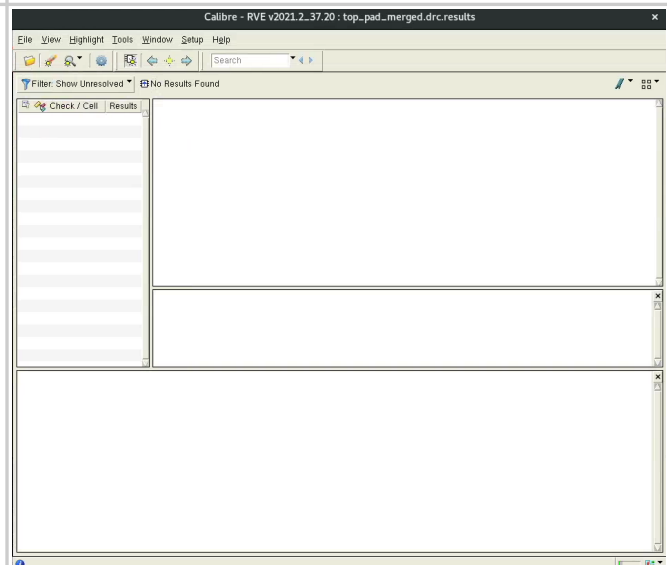


Fig. 6: The clean DRC: full-chip

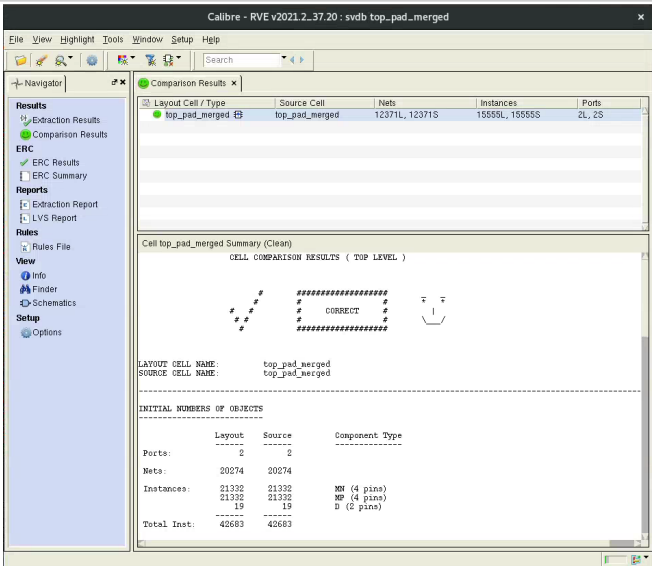


Fig. 7: The clean LVS: full-chip

Parameter	Value
Technology	TSMC 180nm 1P6M CMOS
Supply Voltage	1.8V
Core Area	0.104 mm ²
Total Area	2.607 mm ² (with pads)
Gate Count	4,524 gates
Flip-Flops	536
Metal Layers	6 (M1-M6)

Fig. S1: DES engine summary

Parameter	Value
Operating Frequency	33.3 MHz
Throughput	133 Mbps
Total Power	8.87 mW
Dynamic Power	8.87 mW (99.997%)
Static Power	0.29 μW (0.003%)
Energy/Block	4.26 nJ
Energy/Bit	66.7 pJ

Fig. S2: Performance and power characteristics

Component	Power (mW)
Registers (536 FFs)	4.013
Combinational Logic	3.197
I/O Pads	1.337
Clock Tree	0.319
Total	8.866

Fig. S3: Power distribution